

Mapping Quantum Reservoir Advantage

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Anonymous submission

Abstract—Quantum reservoir computing is often evaluated as a direct forecasting competitor to classical reservoirs. For practitioners, the sharper question is conditional: which datasets make a quantum reservoir worth testing? We present a dataset-regime benchmark comparing globally frozen QRC protocols against a feature-matched echo state network. Each dataset is described by measured time-series properties, and a meta-model maps those properties to empirical QRC usefulness. The result is not broad average superiority. QRC advantage is selective and concentrated in slow, stateful regimes involving persistence, drift, low-frequency structure, and correlated fluctuations. The benchmark reframes quantum reservoir advantage as dataset-level triage.

Index Terms—quantum reservoir computing, echo state networks, time-series forecasting, benchmark atlas, quantum advantage triage

I. INTRODUCTION

Quantum reservoir computing follows a simple hybrid-learning pattern: fixed quantum dynamics transform an input stream into measured expectation-value features, and only a classical readout is trained [1]–[3]. This makes QRC attractive for near-term time-series forecasting because the quantum component is not variationally optimized. The open benchmark question is different from the hardware-scaling question: *given a new dataset, when is a quantum reservoir worth trying against a strong classical reservoir?*

We answer this question empirically through a large dataset-regime atlas. We use “quantum reservoir advantage” in a deliberately scoped sense: a frozen QRC protocol has lower test error than a frozen feature-matched ESN on the same forecasting task. This is not a complexity-theoretic quantum-advantage claim. It is a benchmark claim about which dataset regimes favor a fixed quantum reservoir feature map over a canonical ESN feature map.

The central result is selective rather than universal. Across the held-out validation split, the best frozen QRC variant beats the ESN in 32.4% of datasets and is clearly useful in 12.5%. The mean best-QRC advantage is negative, so the ESN remains the stronger default. Yet the QRC wins are not noise. They recur across independent discovery and validation splits and are predictable from precomputed dataset descriptors. A meta-model trained on discovery rows distinguishes QRC-useful validation cases with ROC-AUC 0.836 and PR-AUC 0.431, more than three times the 12.5% base useful rate.

This changes the practical question. Instead of asking whether QRC is generally better than ESNs, the atlas asks whether a dataset lies in a QRC-favorable regime. The strongest regimes involve slow, stateful temporal structure: drifting trends, regime changes, persistent low-frequency components,

long memory, and temporally correlated fluctuations. The useful regions are not summarized by one scalar descriptor. They arise from interactions among trend strength, spectral structure, persistence, volatility memory, predictability, and nonstationarity.

Our contributions are:

- **A frozen-protocol QRC–ESN atlas:** we evaluate 20,000 labeled discovery/validation datasets from 52 base generators and 16 broad time-series families under globally calibrated, then frozen, QRC and ESN protocols.
- **Dataset-level advantage triage:** measured dataset properties predict where QRC is useful, enabling a practitioner to screen candidate datasets before running a quantum reservoir.
- **Regime-level evidence:** QRC-favorable regions concentrate in slow, stateful dynamics rather than in a broad average advantage.
- **Anti-circularity checks:** the prediction signal remains strong after removing direct predictability proxies and under restricted feature families.
- **Claim boundary:** we provide a benchmark and regime map, not a proof of computational quantum advantage or a causal mechanism for quantum dynamics.

II. PRELIMINARIES

We formalize QRC and ESN forecasting as fixed feature-map comparisons with a trained readout. For a univariate or input-driven time series, a reservoir maps the history up to time t into a feature vector $\Phi_t \in \mathbb{R}^D$. The readout is fit on a training split and evaluated on a held-out test split. The primary error metric is normalized root mean squared error,

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{T} \sum_t (y_t - \hat{y}_t)^2}}{\text{std}(y_t)}. \quad (1)$$

A. QRC as a fixed measured feature map

For QRC, the feature map is generated by fixed quantum dynamics. Given input u_t , the reservoir state evolves under an input-dependent quantum channel

$$\rho_t = \mathcal{Q}_{u_t}(\rho_{t-1}), \quad \Phi_t = (\text{Tr}[O_a \rho_t])_{a=1}^D, \quad (2)$$

where O_a are measured observables. Only the classical readout is trained. In the v5 protocol, the QRC family consists of three canonical frozen variants: a mechanistic disordered spin-QRC (QRC-M), an encoding-enhanced variant with fixed reuploading (QRC-E), and a mildly dissipative variant (QRC-D). All use the same feature dimension as the ESN comparator.

B. ESN comparator

Echo state networks are the classical reservoir-computing baseline most directly matched to QRC: random recurrent dynamics produce a nonlinear state, and only the readout is trained [4], [5]. Our ESN is a sparse random leaky reservoir with globally calibrated spectral radius, leak rate, and input scale. Its reservoir size is matched to the QRC feature dimension. This makes the comparison a fixed quantum feature map versus a fixed classical recurrent feature map, not a per-dataset tuning contest.

C. Dataset-regime view

The atlas treats QRC advantage as a property of a dataset regime. For each dataset i , define

$$A_i = \text{NRMSE}_{\text{ESN},i} - \min_{v \in \{M,E,D\}} \text{NRMSE}_{\text{QRC},v,i}. \quad (3)$$

Positive A_i means that the best frozen QRC variant beats the frozen ESN. We call a case *useful* when $A_i \geq 0.05$. The meta-model then asks whether measured descriptors z_i predict the label $A_i \geq 0.05$ before running the QRC.

III. RELATED WORK

Reservoir computing was introduced to exploit high-dimensional recurrent dynamics while training only a simple readout [4]–[6]. ESNs remain the most direct classical comparator for QRC because they share the same learning structure: fixed recurrent dynamics, nonlinear temporal features, and a trained linear readout. NVAR provides an important fixed-feature robustness baseline for nonlinear forecasting, but it is not the closest reservoir-to-reservoir comparison [7].

QRC transfers this architecture to quantum dynamics. Disordered spin ensembles, spatial multiplexing, and driven quantum systems have been proposed as fixed temporal feature maps [1]–[3]. More recent work shows why protocol details matter: spin-network performance depends on dynamical regimes [8], dissipation can be an active resource rather than a nuisance [9], and input encoding can itself provide strong nonlinear transformations [10]. These results motivate our split between mechanistic, encoding-enhanced, and dissipative QRC variants.

Time-series benchmark archives and forecasting competitions show that model ranking is strongly dataset-dependent: seasonality, persistence, noise, nonstationarity, and regime changes can dominate average scores [11], [12]. Our contribution is to connect this dataset-regime view to QRC. Rather than reporting only a mean QRC-versus-ESN score, we measure dataset properties, label empirical advantage under a frozen protocol, and learn where in property space QRC becomes useful.

The paper is therefore complementary to mechanism studies. It identifies *where* QRC is empirically worth testing under a controlled benchmark. It does not prove *why* those regimes arise from a particular quantum mechanism.

TABLE I

FROZEN RESERVOIRS USED IN THE BENCHMARK. ALL RESERVOIRS ARE CALIBRATED ONCE, THEN REUSED WITHOUT PER-DATASET TUNING.

Model	Role in the paper	Dim.
QRC-M	Mechanistic spin-QRC with input injection only. Tests the fixed coupled reservoir dynamics with minimal extra encoding.	63
QRC-E	Same spin reservoir with fixed repeated input reuploading. Tests a practical encoding-enhanced QRC feature map.	63
QRC-D	Same spin reservoir with mild dissipation. Tests a realistic open-system QRC design axis.	63
ESN	Sparse random leaky ESN with globally calibrated spectral radius, leak, and input scale. Primary classical comparator.	63

IV. METHODOLOGY

The benchmark is designed to avoid two common confounds: per-dataset reservoir tuning and post-hoc model selection. We calibrate QRC and ESN hyperparameters once on held-out synthetic calibration datasets, then freeze them for the atlas evaluation.

A. Frozen QRC variants

All QRC variants use a six-qubit disordered transverse-field spin reservoir with complete coupling topology, five internal layers, and three virtual nodes, giving 63 measured features. QRC-M uses input injection without repeated reuploading or dissipation. QRC-E uses the same reservoir with fixed input reuploading, motivated by practical QRC feature-map protocols and by the need to expose encoding contributions explicitly [10]. QRC-D uses the same reservoir with mild local dissipation, reflecting the current view that dissipation can be a useful design axis in QRC [9]. Calibration selects one global setting per variant: QRC-M uses $J = 1.5, dt = 0.25$, QRC-E uses $J = 1.0, dt = 0.25$, and QRC-D uses $J = 1.0, dt = 0.15$ with mild damping.

B. Frozen ESN comparator

The ESN is a canonical sparse random leaky reservoir with the same feature dimension as QRC. The global calibration selects spectral radius 0.7, leak 0.3, and input scale 0.3. Once selected, the ESN is frozen for every atlas row. This ensures that the benchmark measures dataset-regime sensitivity rather than per-dataset reservoir optimization.

The ESN is intentionally the primary comparator because it is the closest classical analogue of QRC: both models use fixed recurrent dynamics and train only a linear readout. Linear ridge, gradient boosting, and NVAR are retained as contextual baselines, but the headline label is always QRC versus ESN. This keeps the title claim narrow: the paper maps empirical QRC advantage relative to the traditional reservoir-computing baseline, not relative to every possible forecasting method.

C. Dataset descriptors and meta-model

Each dataset is profiled by 30 Tier-A descriptors. The original 20 core descriptors include autocorrelation time, mutual-information minimum, memory capacity, linear predictability, nonlinear-gain contrast, signal-to-noise ratio, Lyapunov and 0–1 chaos diagnostics, spectral entropy, dominant frequency,

stationarity tests, differencing order, scaling exponents, permutation/sample entropy, Hurst estimate, forecastability, and GBM predictability. Ten additional descriptors capture predictability gaps, volatility memory, ARCH structure, recurrence, spectral slope/centroid, trend strength, changepoints, and Lempel–Ziv complexity.

The descriptors are organized into six interpretable blocks: memory and persistence; predictability; spectrum and oscillation; nonstationarity and drift; noise and volatility; and nonlinear, chaotic, entropy, and recurrence structure. The blocks are intentionally redundant at the measurement level. For example, slow structure can appear as a high autocorrelation timescale, a large scaling exponent, low spectral centroid, strong trend score, or a stationarity-test signal. The meta-model is allowed to combine these partially overlapping views, while the ablation suite checks that the result is not carried by one proxy family.

The meta-model is trained only on discovery rows and evaluated once on validation rows. It uses gradient-boosted trees for regression on A_i and classification of $A_i \geq 0.05$. We report ROC-AUC and PR-AUC because useful QRC cases are rare. Feature-set ablations test whether the signal survives after removing direct predictability proxies or restricting to persistence/spectral/nonstationarity descriptors.

D. Quantum-advantage triage

The output is a triage map, not a universal model ranking. Given a new dataset, the practitioner measures the descriptors z_i , locates the dataset in the atlas, and estimates whether QRC is worth testing. We call this *dataset-level quantum-advantage triage*: identifying plausible empirical QRC advantage before investing in quantum-reservoir evaluation.

V. EXPERIMENTAL SETUP

The v5 atlas contains 20,000 labeled datasets split into 10,000 discovery and 10,000 validation rows. They are selected from a 50,000-row candidate pool and span 52 base generators grouped into 16 broad families. The families include chaotic maps and flows, input-driven memory systems, oscillatory and quasiperiodic systems, long-range processes, colored noise, drifting trends, regime switching, seasonal/calendar structure, volatility-heteroskedastic processes, delay dynamics, intermittent sparse processes, nonlinear autoregressions, heavy-tail jumps, and multiscale composites.

The 50,000-row candidate pool is used to make the final atlas broad without turning the paper into an unstructured random sweep. Candidate series are first generated across the full taxonomy with varied seeds and parameters. The final discovery and validation rows are then selected to cover both broad family balance and frontier enrichment near the regions where QRC and ESN are close. This matters scientifically: a purely average benchmark would mostly estimate that ESN is the stronger default, whereas a regime atlas must also contain enough near-boundary and QRC-favorable examples to learn where the boundary lies.

TABLE II
FROZEN v5 PROTOCOL AND HELD-OUT VALIDATION SUMMARY.

Item	Value
Discovery / validation rows	10,000 / 10,000
Families / base generators	16 / 52
Measured descriptors	30 Tier-A properties
QRC variants	QRC-M, QRC-E, QRC-D
Feature dimension	63 for QRC and ESN
Best-QRC win rate	32.4% on validation
Best-QRC useful rate	12.5% on validation
Mean best-QRC advantage	−0.058 on validation

TABLE III
DESCRIPTOR BLOCKS USED FOR THE 30-FEATURE REGIME MAP.

Block	Examples
Memory and persistence	Autocorrelation time, memory capacity, DFA exponent, Hurst estimate
Predictability	Linear and GBM NRMSE, nonlinear-gain contrast, predictability gap
Spectrum	Spectral entropy, flatness, dominant frequency, spectral slope and centroid
Drift and stationarity	ADF/KPSS signals, differencing order, trend strength, change-points
Noise and volatility	SNR, volatility autocorrelation, ARCH statistic
Nonlinear structure	Lyapunov estimate, 0–1 statistic, entropy, recurrence, Lempel–Ziv complexity

The target is one-step or short-horizon forecasting with train/validation/test splits derived from each generated series. All preprocessing is fit on training data only. The v5 evaluation records NRMSE and NMAE for linear ridge, gradient boosting, NVAR, the frozen ESN, and all three QRC variants. The primary label is the NRMSE advantage of the best QRC variant over the ESN. We also retain variant-specific advantages so that the atlas can distinguish whether QRC-M, QRC-E, or QRC-D is responsible for a QRC-favorable row.

NRMSE is the primary metric because it is scale-free and standard for forecasting comparisons. NMAE is recorded as a robustness metric because it is less sensitive to rare large errors. The main paper uses NRMSE labels; the artifact tables preserve NMAE columns for follow-up robustness checks and for readers who prefer absolute-error normalization.

The paper-facing analysis package writes all tables and figures from the completed v5 outputs. The main validation outputs are the complete validation table, the split/family/variant summaries, the discovery-trained validation predictions, the feature-set robustness table, and the rule-pocket table. All reported numbers in this paper are taken from these generated artifacts.

VI. EVALUATION

The evaluation is organized around the five paper claims: selectivity, predictability, slow-stateful regimes, interacting descriptors, and dataset-level triage.

A. QRC is selective, not broadly dominant

QRC does not win on average. In validation, the mean ESN NRMSE is 0.805 and the mean best-QRC NRMSE is 0.863, giving mean best-QRC advantage −0.058. However, QRC wins

V5 regime map

All validation datapoints projected from the 30 Tier-A descriptors; outlined points are useful QRC cases.



Fig. 1. Validation regime map. Each point is one synthetic time-series dataset projected from 30 measured descriptors. Color denotes best-QRC advantage over the frozen ESN; outlined points meet the useful threshold. The side panels summarize the families where QRC is most often useful and which QRC variant is selected.

TABLE IV

VALIDATION PERFORMANCE BY FROZEN MODEL. ADVANTAGE IS MEASURED AGAINST THE FROZEN ESN.

Model	Mean NRMSE	Median	Win	Mean adv.
Linear ridge	0.802	0.945	47.5	+0.003
GBM	0.782	0.907	39.3	+0.023
NVAR	0.844	0.962	35.3	-0.039
Frozen ESN	0.805	0.919	—	0.000
Best QRC	0.863	0.959	32.4	-0.058

TABLE V

VALIDATION PERFORMANCE BY FROZEN QRC VARIANT.

Variant	Mean adv.	Median adv.	Win	Useful
QRC-M dynamics	-0.100	-0.029	22.0	9.4
QRC-E reupload	-0.155	-0.058	16.2	6.9
QRC-D dissipative	-0.239	-0.090	14.0	5.5
Best QRC	-0.058	-0.011	32.4	12.5

are not random leftovers: the best frozen QRC variant beats the ESN in 32.4% of validation rows and reaches the useful threshold in 12.5%. Discovery is similar, with 32.6% win rate and 11.8% useful rate. This discovery/validation agreement supports the first claim: empirical quantum reservoir advantage is selective but reproducible.

The best-QRC row is not per-dataset tuning. It is the best member of a small, fixed menu of canonical QRC protocols. The interpretation is practical: if a user is willing to test a few defensible QRC mechanisms, the atlas asks where at least one is useful against the same frozen ESN.

B. Dataset properties predict QRC usefulness

The discovery-trained meta-model predicts validation advantage with $R^2 = 0.407$ and MAE 0.101. For the binary useful label, it obtains ROC-AUC 0.836 and PR-AUC 0.431 against a 12.5% base rate. This is strong enough for triage: the model ranks held-out datasets by QRC usefulness from measured descriptors, before running QRC on those rows.

The important descriptors are not a single axis. Top features include GBM predictability, scaling exponent, predictability-gap structure, spectral centroid, volatility autocorrelation, trend strength, nonlinear-gain contrast, Lyapunov estimate, spectral slope, autocorrelation time, stationarity signal, and SNR. The signs are mixed because the map is interaction-driven: the same descriptor can be favorable in one region and unfavorable in another.

C. Slow, stateful regimes are most favorable

Fig. 3 and Tab. VI summarize the family-level structure. The strongest family is the drifting-trend family, with 47.9% useful rate, 51.8% win rate, and mean advantage +0.240 with 95% bootstrap interval [+0.203, +0.282]. Regime switching and colored noise also have positive mean effects and useful rates above 23%. Intermittent sparse and linear stochastic families have high win rates but small mean effects, indicating many near-ties rather than large QRC gains.

The strongest extracted validation rule pocket is also interpretable. Rows with high trend strength and low spectral centroid form a pocket of 420 validation datasets with 47.4%

V5 meta-model validation

The meta-model is trained on discovery rows and evaluated once on the held-out validation split.

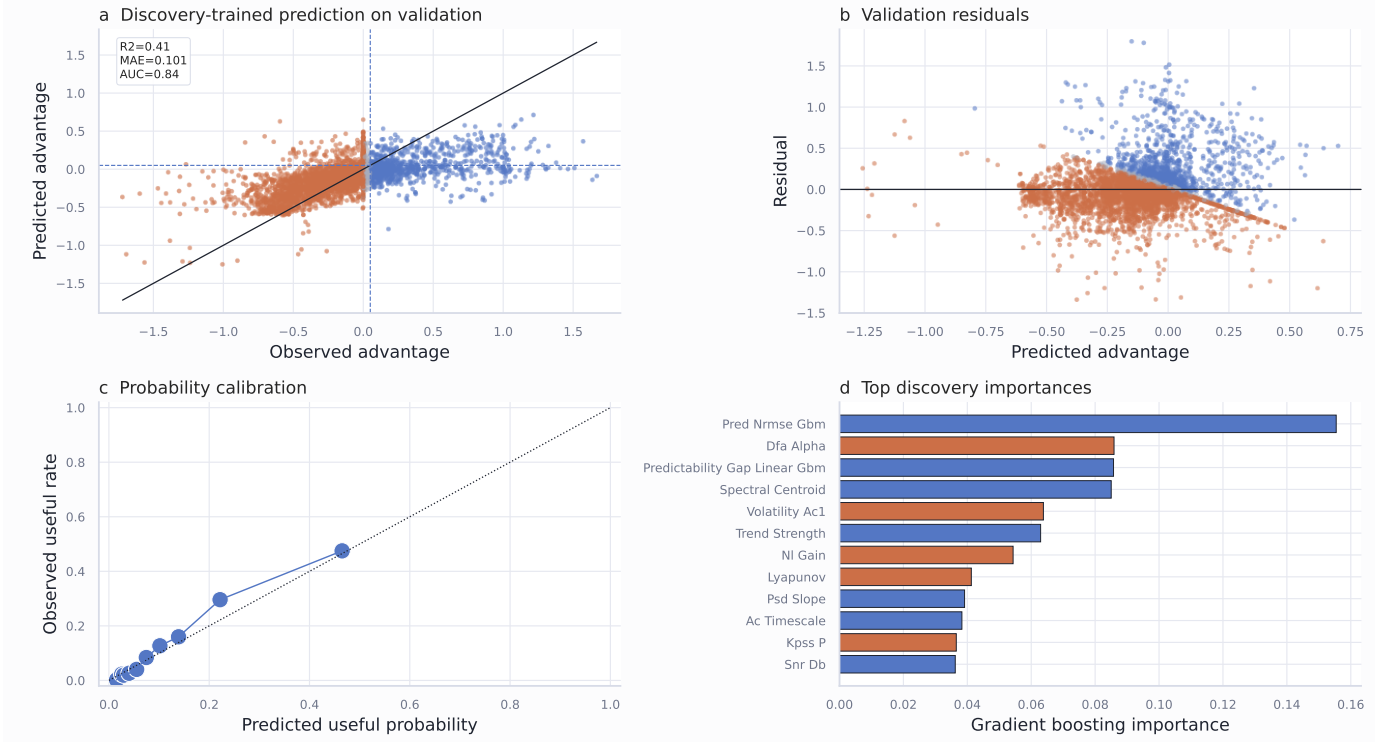


Fig. 2. Meta-model validation. The model is trained on discovery rows and evaluated once on held-out validation rows. The panels show predicted versus observed advantage, probability calibration, and discovery-trained feature importance.

Family-level v5 effects

Family summaries are validation-only; intervals bootstrap rows within each family.

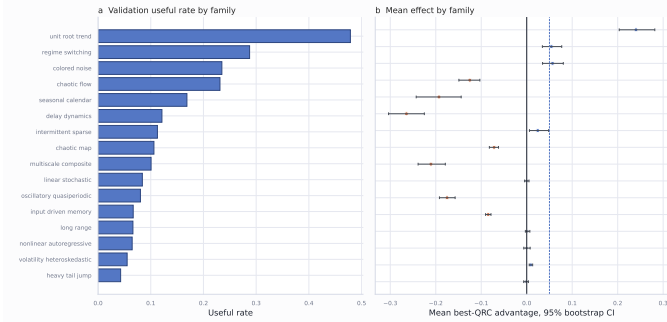


Fig. 3. Validation family effects. Drifting-trend, regime-switching, and colored-noise families have the clearest positive mean effects. Several families contain useful pockets despite negative family-average advantage, showing why row-level triage is needed.

useful rate and mean advantage +0.209. This is not a mechanism proof. It is a compact triage rule: slow, trend-heavy, low-frequency structure is a high-priority region for testing QRC.

D. The map is robust to feature choices

The feature-set robustness table supports the anti-circularity claim. The full 30-feature map reaches validation ROC-AUC 0.836 and PR-AUC 0.431. Using only the original 20 core

TABLE VI
TOP VALIDATION FAMILIES BY USEFUL QRC RATE.

Family	n	Useful	Win	Mean adv.
Drifting trend	353	47.9	51.8	+0.240
Regime switching	313	28.8	48.2	+0.054
Colored noise	315	23.5	48.6	+0.057
Chaotic flow	1137	23.1	29.6	-0.125
Seasonal/calendar	255	16.9	25.9	-0.193
Delay dynamics	231	12.1	15.6	-0.264
Intermittent sparse	142	11.3	53.5	+0.024
Chaotic map	2056	10.6	28.5	-0.072

descriptors gives ROC-AUC 0.828 and PR-AUC 0.423. Removing direct predictability proxies gives ROC-AUC 0.834 and PR-AUC 0.432. Restricting to persistence, spectrum, volatility, and nonstationarity gives ROC-AUC 0.827 and PR-AUC 0.409. Thus the atlas is not simply rediscovering which baseline already won.

Threshold robustness leads to the same conclusion. The validation useful rate is 35.4% at threshold 0, 17.5% at 0.025, 12.5% at 0.05, and 8.5% at 0.1. Stricter thresholds make QRC advantage rarer, but they do not erase the slow-stateful pocket.

E. What the triage map means

The practical output is a screening rule, not a universal deployment rule. A practitioner can measure the 30 descriptors, locate the dataset in property space, and decide whether QRC

TABLE VII
VALIDATION ROBUSTNESS OF META-MODEL FEATURE SETS.

Feature set	R^2	ROC	PR
Full 30 descriptors	.407	.836	.431
Original 20 core	.383	.828	.423
No predictability proxies	.392	.834	.432
No chaos/nonlinear/complexity	.384	.831	.430
Persistence/spectrum/drift	.364	.827	.409
Chaos/nonlinear/complexity only	.324	.798	.359

evaluation is worth the extra cost. High predicted usefulness does not guarantee a QRC win; it identifies a region where the empirical evidence says QRC deserves testing. Low predicted usefulness is equally useful: it says the ESN is likely to be the better default unless there is a separate hardware, physical, or scientific reason to test a quantum reservoir.

VII. CONCLUSION

We introduced a dataset-regime benchmark for empirical quantum reservoir advantage against echo state networks. The main result is deliberately two-sided. ESNs remain the stronger default: QRC does not win on average under the globally frozen protocol. At the same time, QRC wins are reproducible, structured, and predictable from measured dataset properties. The strongest QRC-favorable regimes involve slow, stateful dynamics: drifting trends, regime changes, persistent low-frequency structure, long memory, and temporally correlated fluctuations.

The benchmark therefore turns QRC selection into a measurable triage problem. Practitioners need not ask whether QRC is universally better. They can profile a dataset, compare it with the atlas, and decide whether quantum-reservoir evaluation is worth the cost relative to a strong ESN baseline. This is the practical meaning of quantum reservoir advantage in this paper.

The next scientific step is mechanistic. The atlas identifies where QRC helps; it does not yet prove why. Follow-up work should measure reservoir-state memory kernels, latent-state recoverability, phase/coupling/dissipation ablations, and matched long-memory ESN controls in the high-usefulness pockets identified here. Those studies can test whether the observed slow-structure regimes arise from genuinely quantum feature dynamics or from a broader reservoir-memory design principle.

The strongest claim supported now is therefore not universal quantum advantage. It is a regime-map claim: empirical quantum reservoir advantage is selective, measurable, and predictable under a fair frozen comparison against an ESN. That claim is useful even when QRC loses on average, because it gives the field a concrete way to decide where QRC evidence should be concentrated next.

REPRODUCIBILITY AND CLAIM BOUNDARY

The artifact tree separates the protocol, labels, analysis tables, figures, and paper draft. The main v5 evidence used here is stored under `results_frontier_v5_discovery/`, `results_frontier_v5_validation/`, and `results_v5_publication/`. The title uses “quantum

reservoir advantage” in the empirical benchmark sense: lower forecasting error for a frozen QRC variant than for a frozen feature-matched ESN under the declared protocol. We do not claim computational quantum advantage, broad average QRC superiority, or a proven coupling/interference mechanism.

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